

AI Course

Capstone Project Final Report

For students (instructor review required)

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Bayan: On-Device Arabic Text Simplification for Readers with Dyslexia

تبسيط · Simplify

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Abstract. Arabic poses problems for readers with dyslexia that tools built for Latin scripts do not address: short vowels go unwritten, clitics fuse into long words, and clauses chain across whole paragraphs. We present **Bayan**, an Android system that rewrites a selection of Arabic text, from inside any application and entirely on the phone, into a form that is easier to decode. We first show that the standard evaluation misleads for this task: on SAMER, the only human-written Arabic simplification corpus, copying the input scores 77.5 SARI, and a fine-tuned AraT5v2 that beats copying on the sentences a human changed still loses to it overall. We then train two encoder–decoder models, AraT5v2 (368M parameters) and AraBART (139M), on a 22,733-pair corpus in which every unchanged pair is kept, LLM-written pairs enter only through a meaning filter derived from human rewrites, and a strength tag conditions each source. Building that corpus exposed that 9.3% of a public Arabic simplification dataset overlaps the standard test sets. To measure what a reader needs rather than agreement with one editor, we introduce **BayanBench**, 1,981 items split by document, which scores meaning with an open model (Gemma 4 31B) and simplification, restraint and correctness with code, each with cluster-bootstrap intervals. Five raters and a reader with dyslexia rated 290 outputs and comparisons; the meaning scorer separates the outputs they judged faithful from the rest with an AUC of 0.85 (0.91 when a minor change counts as faithful). On the benchmark, meaning and simplification trade off: the medium strength setting cuts 6.4 words from the longest clause and keeps the meaning in 62.5% of outputs (AraT5v2), while the shipped int8 AraT5v2 bundle keeps it in 78.4% and cuts 1.4. AraT5v2 keeps meaning 22.2 points more often than AraBART without a tag, and no system yet leaves easy text alone. The reader with dyslexia found 17 of 30 rewrites easier to read. On a Snapdragon 870 phone the first word of a rewrite appears 82–172 ms after the request with the int8 AraBART bundle (222 MB) and 177–318 ms with AraT5v2 (471 MB), and each later sentence starts within 30 and 85 ms.

1. Introduction

1.1. Background Information

Developmental dyslexia affects an estimated 11% of Arab primary-school children, pooled across 18 studies ($N = 30,243$) [1]. It impairs reading through phonological processing, and Modern Standard Arabic adds obstacles of its own. Short vowels are normally left unwritten, so one written form carries several readings: **كتب** is *kataba* (he wrote), *kutiba* (it was written) or *kutub* (books), and the reader must resolve it from context. Clitics attach directly to the stem, producing long words that resist chunking, and Arabic-speaking children with reading disabilities show early deficits in exactly this morphological layer [2]. Prose chains clauses with **و** and **ف** across whole paragraphs, so one sentence can exceed working memory.

The evidence on what helps is uneven, and we build on the parts that have been measured. More frequent and shorter words sped reading and improved comprehension for readers with dyslexia in eye-tracking studies [3, 4]. Unpacking bound morphemes into separate words helped Arabic readers with reading disabilities and not their typical peers, the one Arabic finding with a tested group interaction [5]. Texts simplified by rules that split subordinate and coordinated clauses into short sentences were read faster and understood better by beginning readers, and most by the weakest of them [6]. Full diacritization, by contrast, is contested: it adds visual crowding [7], and its benefit for our readers has not been tested. We therefore treat *simplify first* as the core intervention and diacritization as an optional later stage.

1.2. Motivation and Objective

Readers meet difficult Arabic inside other applications — a message, a news page, a PDF — not in a reading app. Bayan therefore appears where the text already is: selecting text on an Android phone offers تبسيط (*Simplify*) in the system selection menu, and the rewrite opens in a panel over the same app. Every model runs on the phone, so the reader’s text never leaves it and the tool works offline.

The closest prior systems leave this setting open. Dyslexia-targeted neural simplification exists for English [8]; Arabic simplification corpora and models exist without a dyslexia target [10, 12]; and the one Arabic dyslexia study with a language model used it as a tutor, not as an automatic rewriter [9]. Our literature searches of 17 and 23 September found no system that combines the three: Arabic, a dyslexia-oriented rewrite, and on-device inference. Building one raises three questions, which organise this report:

- RQ1** Does the standard reference-based evaluation measure what this task needs, and what does a model learn from the only human-written corpus?
- RQ2** Can one model, trained on meaning-filtered mixed-domain data with a strength tag, make Arabic text measurably easier to decode without losing its content?
- RQ3** Does it run on a mid-range phone at interactive latency, and what do quantisation and the deployed input format cost?

We set four objectives against which subsection 4.1 reports the outcome:

- O1** A simplifier that beats copying the input on held-out sentences a human simplified, and that makes text measurably easier without losing its meaning.
- O2** Diacritics restored on the simplified text without altering its letters.
- O3** The pipeline running on a mid-range Android phone, within 250 MB for the simplifier.
- O4** Evidence from readers with dyslexia, collected with informed consent.

Contributions.

- **An evaluation audit.** We show that copy-heavy test sets make SARI reward inaction, report every score beside a copy baseline and split by whether a human changed the sentence, and document that 9.3% of a public Arabic simplification corpus overlaps the SAMER and BAREC test sets (subsection 3.1, subsection 3.3).
- **A training recipe for one model at several strengths**, built from that audit: all unchanged pairs kept, LLM-written pairs admitted by a meaning filter calibrated on human rewrites, and a strength tag that becomes the reader’s setting (subsection 2.2).
- **BayanBench**, a benchmark for dyslexia-oriented Arabic rewriting that reports meaning, simplification, restraint and correctness as separate measures, with an open meaning scorer that anyone can rerun, document-level splits and cluster-bootstrap intervals (subsection 2.5).
- **A human evaluation** of 290 rating tasks by five raters and a reader with dyslexia, which validates the meaning scorer against people and rates every system on meaning, ease and correct Arabic (subsection 3.4).
- **An on-device system**, with int8 encoder–decoder bundles decoded token by token inside an Android app and measured on a phone (subsection 2.4, subsection 3.6).

subsection 1.3 places the work in the literature. Section 2 describes the data, the models, the system, the benchmark and the rating protocol; Section 3 reports the results for each research question; Section 4 draws the conclusions and states the limitations.

1.3. Related Work

Arabic simplification data. SAMER [10] is the only human-written multi-level Arabic simplification corpus: 15 novels rewritten at two easier levels, each version keeping the original's sentences by design. BAREC [11] labels 69,441 sentences with 19 human reading levels but is not parallel. Baseet [12] provides three LLM-written levels per source, and DAASI [13] human-verified rewrites of government and insurance text. No corpus targets readers with dyslexia, and none teaches sentence splitting: a sentence is added in 0% of SAMER pairs, 2–4% of Baseet's and 8% of DAASI's (subsection 3.2).

Controllable simplification. A control token prepended to the source lets one model rewrite for different audiences, first with target-grade tokens [15] and later with measured attributes such as length and lexical complexity [16, 17]. Our strength tags follow this line with one difference: each tag names the corpus style it was learned from, so that a meaning filter can be applied to exactly the sources that need one.

Evaluating simplification. SARI [18] is the field's standard, but it correlates only $r = 0.36$ with human judgements of simplicity [20], and similarity-based metrics detect deleted content but not inserted or substituted content [21]. Reference-free estimates of the reading level track simplicity better [22], and edit-level error taxonomies track meaning [23]. For benchmark design we follow behavioural testing [24], construct-validity and reporting checklists [25, 26], and document-clustered intervals [27]. LLM judges are known to favour outputs close to data they wrote or approved [28], so an automatic judge must be checked against people before its scores are trusted; we report agreement between raters with Krippendorff's α [29].

1.4. Members and Role Assignments

Every member merged work into the team repository; the evidence column cites the pull requests and issues that hold it (#).

1.5. Schedule and Milestones

The work reported here ran from 13 September to 3 October 2026, ahead of submission on 5 October and the pitch on 6 October. It went through two complete build-and-measure cycles. Model 1 was trained, scored on locked test sets and diagnosed. Model 2 was designed from that diagnosis, trained on two architectures, exported to the phone, tested on a device, and scored on a benchmark built for the task. The last week replaced the benchmark's closed meaning judge with an open scorer and checked that scorer against human ratings. Figure 1 shows the timeline; Table 2 compares target and actual dates.

Table 1. Members, roles, and the work each delivered.

Member	Role	Delivered (evidence)
Marwan Elamami	Team lead; integration; on-device model	Product-form decision (#9); evaluation protocol and fixes (#16, #18); model 1 diagnosis; Baseet acquisition and leakage filtering; model-2 recipe, data and task specifications (#27–#34); on-device engine and app; phone test; BayanBench and its v2 package (#48); rating site and analysis; Action Plan (#25); this report
Abdulrahman Khengari	Data lead	Validators for generated pairs (#10, #11); CAMEL readability check (#17, #20); synthetic corpus of 13,072 pairs, split and published (#8, #38, #39); BayanBench v2 design, open meaning scorer runs and rating-set design (#48); pitch decks (#50)
Ahmed Alaeb	Model training	SAMER conversion; model 1 trained and documented (#5, #22); model 2 on AraT5v2 trained and published (#28, #40); human ratings (#48)
Sanad Ali	Diacritization; co-training	Four-system diacritization benchmark; Libtashkeel selected (#6, #23); model 2 on AraBART and the speed benchmark (#29, #35); human ratings (#48)
Mohammed Thabet	Android application	PROCESS_TEXT plugin, result sheet and 41 tests (#14, #21); model download pipeline, sheet states and typography settings (#31, #37); human ratings (#48)
Abdul Majid Mraied	Evaluation	Scoring and leakage scripts (#7, #15); locked-set converter, copy baselines, generate-and-score path (#19, #24, #26)

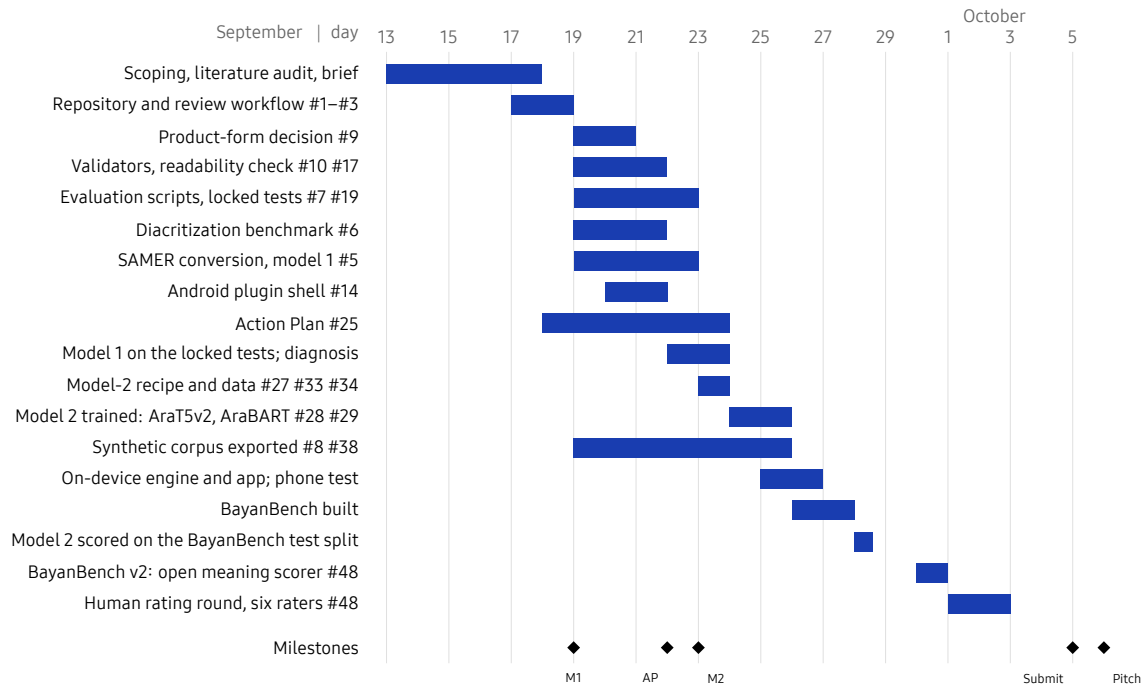


Figure 1. Timeline as executed, 13 September to 3 October. AP = Action Plan; M1, M2 = mentoring sessions. Numbers after a task are its GitHub issues.

Table 2. Milestones: target against actual.

Milestone	Target	Actual
Evaluation scripts ready (#7)	Mon 21 Sep	Mon 21 Sep
Diacritizer selected (#6)	Mon 21 Sep	Mon 21 Sep
Android plugin shell (#14)	—	Mon 21 Sep
Action Plan submitted	Tue 22 Sep	Tue 22 Sep
Model 1 scored on the locked test sets	Wed 23 Sep	Wed 23 Sep
Model-2 data built; tasks issued (#28–#32)	—	Thu 24 Sep
Model 2 trained, both architectures (#28, #29)	Fri 25 Sep	Fri 25 Sep
Synthetic corpus exported (#8, #38)	Fri 25 Sep	Sat 26 Sep
Model running on a phone, latency measured	Fri 2 Oct	Sat 26 Sep
BayanBench built; model 2 scored on its test split	—	Mon 28 Sep
BayanBench v2: open meaning scorer; all systems scored (#48)	—	Wed 30 Sep
Human rating round complete, six raters (#48)	Fri 2 Oct	Sat 3 Oct

2. Project Execution

2.1. Data Acquisition

We acquired five corpora and produced a sixth (Table 3). SAMER is used under written approval from CAMEL Lab to fine-tune and to publish the resulting weights non-commercially; the corpus itself is never redistributed, and no SAMER sentence is reproduced in this report. Baseet is used with its authors' permission. Two test splits, SAMER test and BAREC test, were locked before any training.

Table 3. Corpora. Sizes are rows in the files we hold.

Corpus	Size	Licence	Used for
SAMER [10]	14,343 / 2,983 / 3,277 pairs (train / dev / test)	research; written approval	models 1 and 2; locked test set
BAREC [11]	54,845 / 7,310 / 7,286 sentences	CC BY-SA 4.0	locked test set; BayanBench core items; generation sources
Baseet [12]	34,294 sources, three levels each; 31,104 after filtering	with permission	model 2 (6,888 meaning-filtered pairs)
DAASI [13]	1,509 pairs + 350 held out	CC0	model 2 (1,502 pairs); out-of-domain test
Tashkeela [14]	743 test sentences	open	diacritization benchmark
Bayan synthetic v0	13,072 pairs (12,419 generated, 563 identity, 90 scripture)	CC BY-SA 4.0, published	a released artefact; not used to train model 2

The synthetic corpus. Our generation pipeline takes BAREC train and dev sources, draws five candidates per source from DeepSeek V4.1 Flash, and accepts one only if an LLM equivalence validator and the CAMEL BAREC readability model both pass it. The validator is a DSPy program optimised with MIPROv2 and reaches 84% accuracy on 25 held-out annotated pairs at the chosen threshold. The export is split by a salted hash of the normalised source, so that a sentence repeated under several BAREC identifiers never straddles two splits; a classifier trained to tell the splits apart scores AUC 0.51. BAREC test was never used. The corpus was completed after model 2 had been trained, so no result in this report depends on it.

2.2. Training Methodology

Task. Given a source sentence x and a strength tag t , the model produces a rewrite $\hat{y} = f(t \oplus x)$ that should keep the meaning of x and be easier to decode. Training minimises token-level cross-entropy on pairs $(t \oplus x, y)$; decoding is greedy, the setting the phone runs.

Base models. Simplification is a sequence-to-sequence task, so every candidate is a pre-trained encoder-decoder transformer (Table 4). AraT5v2 came first because it beat AraBART at all three levels in Baseet's head-to-head evaluation; AraBART is the on-device candidate because it is 2.6 times smaller.

Input normalisation. No vocabulary in Table 4 was built for diacritized text: in AraT5v2 only 1.6% of pieces carry a diacritic, and AraBART cannot represent one at all. Every model therefore receives its source with tashkeel and tatweel removed, in training and in the app; subsection 3.3 measures what happens otherwise.

Model 1. We fine-tuned AraT5v2 to rewrite SAMER's level-5 sentences into their level-3 sim-

Table 4. Candidate base models. “Tashkeel” is measured on 1,000 diacritized BAREC dev sentences: vocabulary entries containing a diacritic, and how many more tokens the same sentences cost with diacritics than without.

	AraT5v2-base-1024	AraBART	HPLT T5 base (Arabic)
Family	T5 [35]: span-corruption pretraining, relative position buckets	BART [36]: denoising pretraining, learned positions	T5 in the NorT5 variant [37]
Layers (enc/dec), width	12/12, $d=768$, 12 heads, gated-GELU FFN 2048	6/6, $d=768$, 12 heads, GELU FFN 3072	12/12, $d=768$, 12 heads, FFN 2048
Parameters	368M (46% in two untied $110k \times 768$ embeddings)	139M	$\approx 294M$
Vocabulary	110,208 SentencePiece	50,002 SentencePiece BPE	32,768
Context	1,024 tokens	1,024 positions	512 positions
Pretraining	Arabic MSA and dialects [32, 33]	large Arabic corpora [34]	HPLT v3 web crawl, Arabic subset [38]
Tashkeel	1.6% of pieces; $2.57 \times$ tokens	0 pieces; every mark becomes <unk>; $4.53 \times$ tokens	0 pieces; $1.81 \times$ tokens
Licence	none stated on the model card	Apache-2.0	Apache-2.0
Used in	models 1 and 2	model 2	not trained

plifications with the prefix :بسط (*simplify*): AdamW at learning rate 10^{-4} with weight decay 0.01 and 100 warm-up steps, an effective batch of 16, mixed precision, 10 epochs (3,250 steps) and seed 42. Pairs whose simplification equals the source (42% of SAMER train) were capped at 20% of training, and the checkpoint with the best SARI on a 500-row dev subset was kept.

Model 2. The diagnosis of model 1 (subsection 3.3) identified three problems, and the model-2 recipe changes one thing for each:

- Inaction was rewarded for the wrong reason.* Capping unchanged pairs at 20% while choosing the checkpoint on a dev sample that is 43% unchanged pulls the model both ways. Model 2 keeps all 6,033 unchanged SAMER pairs, so it learns when not to edit, and selects its checkpoint on SARI over dev rows a human changed.
- SAMER is novels, and it teaches word substitution.* Model 2 adds DAASI’s everyday and administrative text and Baseet’s educational text. Baseet’s targets are LLM-written, and many are summaries that drop facts, delete answer options or invent a speaker. A Baseet pair is kept only if it passes a *meaning filter*: LaBSE similarity ≥ 0.646 , a target at least 67% of the source’s length, and every number kept. Each threshold is the 5th percentile of the same measure over DAASI’s human-verified rewrites, so a pair is rejected only when it departs from its source more than 95% of human simplifications do.
- One strength does not fit every reader.* Every source carries a tag naming the style it was learned from (Table 5). At run time the tag is the reader’s strength setting, so one model serves light, medium and strong rewrites [15, 16].

Model-2 training. Both architectures were trained on the same data with one shared script, so that the architecture is the only difference: 10 epochs, batch 8 with 2 gradient-accumulation steps, fp16, maximum length 256, seed 42, evaluation twice per epoch with patience 3, and the checkpoint chosen on SARI over the 600 changed rows of a fixed 900-row SAMER dev sample. AraT5v2 used learning rate 10^{-4} and the prefix :بسط; AraBART used $5 \cdot 10^{-5}$ and :بسط without

Table 5. Model-2 training set (22,733 pairs). For each Baseet source, the mildest level whose target is shorter per sentence or has fewer rare or long words is used. No source overlaps any test set.

Tag	Taught by	Pairs	Meaning at run time
[S0]	SAMER L5→L3, all pairs (8,310 changed, 6,033 unchanged)	14,343	minimal: literary word swaps
[S1]	Baseet level 3, meaning-filtered	3,508	light
[S2]	Baseet level 2, meaning-filtered	1,954	medium (evaluated here)
[S3]	Baseet level 1, meaning-filtered	1,426	strong
[SA]	DAASI train	1,502	everyday and administrative style

the shadda, which its tokenizer maps to an unknown token. The script disables length-grouped batching, which in the pinned transformers 4.46 also shuffles the evaluation set and would score each prediction against another row’s reference; it stops if predictions and references ever fall out of order, and the logs show full alignment at all 40 evaluations.

2.3. Workflow

Each task is a GitHub issue with an owner, a reviewer and a definition of done; changes arrive as reviewed pull requests. Over the period reported, the team opened 23 issues and 28 pull requests, of which 22 were merged, with merged work from every member. SAMER text and model outputs on SAMER move only through private Kaggle datasets and a private Hugging Face organisation, never through the repository.

Figure 2 shows the data path and its three gates. The leakage check compares every training file with both locked test sets, by exact match and by character 5-gram Jaccard overlap of 0.8 or more. The meaning filter admits LLM-written pairs only if they keep the source’s meaning. Generated pairs pass validators before release. The locked test sets are fixed by row count and SHA-256 in a committed manifest and re-verified before every test run.

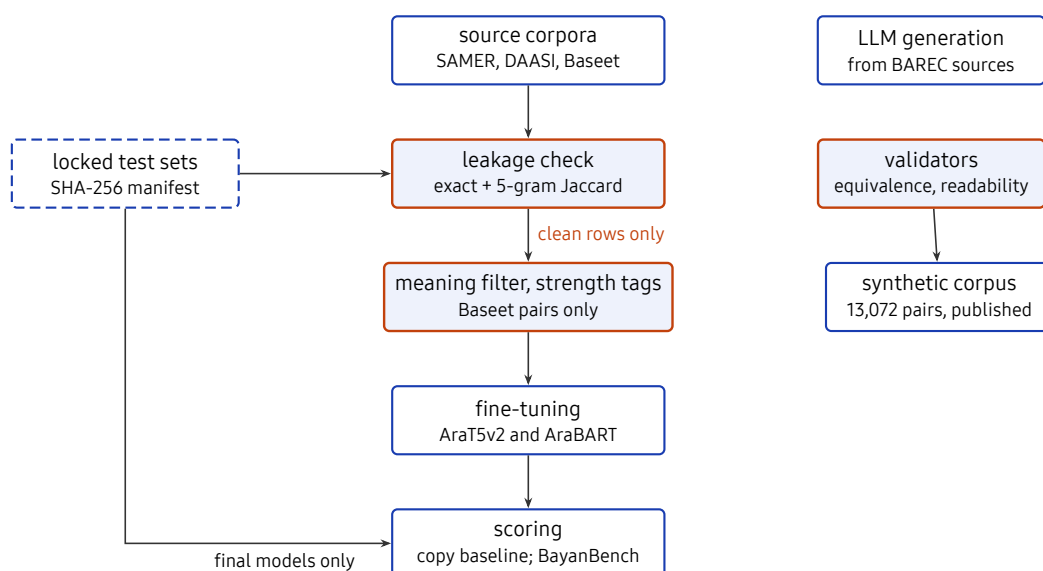


Figure 2. Data and evaluation workflow. Orange boxes are the gates that can reject data. The generated corpus (right) is a separate, published artefact.

2.4. System Design

Figure 3 shows the system as tested on a phone. The Android app registers an activity for `android.intent.action.PROCESS_TEXT` (text/plain, Android 6.0 and later), so تبسيط appears in the selection menu of any app. A translucent activity draws the result over the host app without the overlay permission; an optional floating panel, which does require it, keeps reading while the reader returns to the page.

Inference. The simplifier ships as an int8 ONNX bundle (encoder, decoder with a key-value cache, and tokenizer), downloaded once, verified by SHA-256, and run by ONNX Runtime with a greedy decoding loop written in Kotlin that blocks repeated 3-grams and stops at 256 tokens. The app strips tashkeel, splits the selection into sentences and decodes them one by one, streaming each token to the screen as it is produced. Sentences under six words (headings, labels) pass through unchanged, because the model was trained on longer sentences and invents text around fragments. Each output then passes a *retention fallback*: when a rewrite keeps less than 35% of the source’s words, the source sentence is shown instead. The original is always one tap away.

Read-aloud and diacritization. Voices run on the device through sherpa-onnx. The default, Nabra-7M-Distill, is a Kokoro-architecture Arabic voice exported to int8 (8.7 MB); three Piper voices are optional downloads. The spoken word is highlighted as it is read. The diacritizer, Libtashkeel, was selected on the benchmark in Table 21 but is not part of the build we tested.

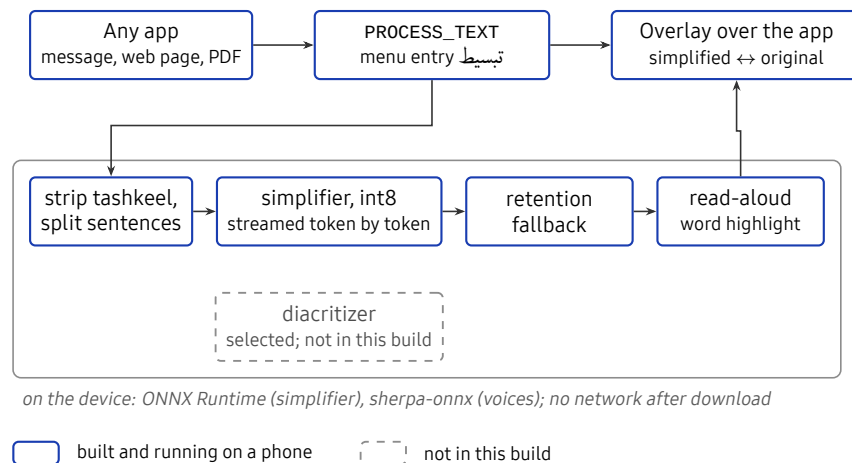


Figure 3. System architecture of the build tested on the phone. The app is entered from the host app’s selection menu and answers with an overlay on top of it.

2.5. BayanBench

SARI measures agreement with one editor’s rewrite. A reader with dyslexia needs something else: that a selection comes back *easier to decode* and *with the same meaning*, and that what should not change does not. No public benchmark tests this for Arabic, so we built one against the construct-validity checklist of [25] and the reporting practices of [26].

Construct. A rewrite is dyslexia-friendly when it says the same thing in a form that is easier to decode and follow: shorter clauses, more familiar words, no new errors, and restraint where the text is already easy. Visual formatting, read-aloud and tashkeel are out of scope, and summarising is always a failure: dropping content never counts as simplification.

Measures. Every output is scored on separate measures, grouped in blocks and never combined into one number (Table 6); the probe tracks follow CheckList [24].

- *Meaning*, from an open model (below): the mean P(same meaning), and *meaning kept*, an output whose P(same meaning) is at least 0.5 and that keeps every number of its source. An unchanged copy keeps its meaning by definition.
- *Simpler*, as changes from source to output, each on the items where there is something to simplify: words cut from the longest clause (sources with a clause over 15 words, BAREC's guideline limit for level 10, about grade 4), reading levels lowered (the maximum sentence level predicted by the CAMEL BAREC readability model, sources at level 10 or above), and hard words removed (words at levels 4–5 of the SAMER readability lexicon).
- *Checks*, deterministic: every number kept, and no house punctuation rule broken.
- *Behaviour*: easy text left unchanged (C), protected text keeping its letters (F), and a cut-off selection keeping its ending (E1).
- *Flags*: a changed count of negations, limits (at least, at most, only) or conditions. A flag marks an output for attention and is never a failure on its own, because a rewrite can rephrase a negation correctly.

A *clause* is a maximal span between punctuation marks (full stop, question and exclamation marks, comma, semicolon, colon, line break), so the clause measure credits added commas as well as split sentences and is read next to meaning. Numbers are compared as multisets after mapping Arabic-Indic and Persian digits to Western ones; negations are counted over eleven negating particles. The pass/fail measures of the first version (no clause over 15 words, at least 60% of the source's words kept) are still computed as diagnostics.

Table 6. BayanBench tracks, with items per track in the dev / test splits.

Track	Behaviour	Measured by	Items
A1–A3	meaning kept: general; hard spots (negation, numbers, conditions, cause, quantity, certainty); high stakes (legal, medical)	meaning scorer; numbers kept; flags	299 / 508
B	long clauses shortened	words cut from the longest clause; meaning scorer	104 / 187
B2, B3	fewer hard words; lower reading level	SAMER word levels; CAMEL BAREC readability model	across items
C	easy text left unharmed	unchanged; meaning scorer	129 / 202
D	correct Arabic	house punctuation rules; human ratings	across items
E1–E4	real selections: cut-offs, messy text, 40+ words, several sentences	end mark; clause length; meaning scorer	139 / 263
F	protected text (Qur'an, Bible, classical poetry)	letters unchanged (never sent to the scorer)	60 / 90
G	on the phone	int8 bundle against the same model in PyTorch; size, time	—
H	people	290 blind rating tasks, six raters (subsection 2.6)	—

Items and splits. The 1,981 items form four sets that are reported side by side and never averaged. *Core* items come from BAREC test documents. *Held-out* items are news and legal text, including 247 BAREC sentences of those domains that no model trained on. *Outside* items are 344 sentences from UN News, WHO fact sheets, newly created Arabic Wikipedia articles, UNHCR help pages and Android help, fetched on 28 September and therefore published after

every corpus we use. *Written* items are 50 medicine-leaflet sentences written for the benchmark by an LLM, kept out of core because they are not real text. Items are split by document (731 dev items from 129 documents, 1,250 test items from 197), so no document contributes to both splits. Any BAREC document with a sentence found in any training or generated data was dropped whole (17 documents), and near-copies were removed by word-trigram overlap.

The meaning scorer. Meaning is scored by Gemma 4 31B [30], run in its 4-bit quantisation-aware checkpoint with vLLM [31], as a scorer rather than a writer: it sees the source and the output, never the system’s name, and answers four yes/no questions (does the rewrite keep exactly the same meaning; does it add a fact; does it drop a fact or qualifier; does it contradict the source). The probability of yes is read from one forward pass, so no text is generated and the score is continuous. Scores are shared data keyed by the content of the (source, output) pair, so scoring a system needs no GPU, an output nobody has scored is reported as pending rather than guessed, and a meaning rate is shown only once 90% of its items are scored. The first version of the benchmark used pass/fail verdicts from a closed model, Gemini 3.1 Pro, with a fixed checklist; the open scorer replaces it because anyone can rerun it, it costs about \$0.12 for 2,700 pairs, and it gives a graded score whose threshold can be checked against people. The 0.5 threshold was set before the human ratings and kept after them (subsection 3.4). Model 2’s training data contains nothing a Gemma model wrote or approved, and the written items come from a different model family, which limits preference leakage [28].

Statistics and release. Every rate and mean carries a 95% interval from a bootstrap over whole documents (2,000 resamples), because sentences from one document pass or fail together [27]; a comparison counts only when the interval of the paired difference excludes zero. The benchmark ships as a Python package whose command-line tool covers the whole cycle (verify, predict, score, meaning-pending, meaning), with its data, model outputs and meaning scores in a checksum-pinned Hugging Face dataset, so any team member can score a model on a laptop, on Kaggle or on Linux.

2.6. Human Rating Protocol

An automatic scorer is only useful if it agrees with people, so we rated a sample of model outputs before trusting it. The rating set holds 245 single outputs and 45 side-by-side comparisons from the dev split (core and outside items, 64 documents), drawn from the six model-2 systems of Table 15. Copies of the input were not rated, because there is nothing to judge, and protected and constructed items (F, E1, E2) were left out. 210 outputs were drawn evenly, 35 per system; 35 more were drawn where the scorer was least sure ($P(\text{same})$ between 0.1 and 0.9, the narrowest band that held enough candidates), so the check does not rest only on easy cases. 80 outputs and 15 comparisons went to every rater, to measure agreement; the rest were split one rating each.

For each output a rater saw the source and the rewrite, without the system’s name, and answered three questions: *meaning* (same, a minor change, a major change), *Arabic* (correct, a minor error, a major error) and *ease* for a reader with dyslexia (easier, the same, harder). For a comparison the question was which version is better for such a reader: easier, with the same meaning. Five team members rated on a web page built for the round, which saved every answer as it was given and bound each rater to one name. A sixth rater, a reader with dyslexia, gave informed consent and answered only the ease question, worded for them as هل كانت أسهل (was it easier for you to read?), on 30 shared outputs and 10 comparisons. An output’s meaning counts as kept when a strict majority of its raters said *same*; a tie counts as not kept.

3. Results

3.1. Data Preprocessing

Conversion. SAMER’s aligned levels were converted to source–target pairs (level 5 → level 3). One converter turns each locked test file into the scoring format, keeps every row (including the 1,599 SAMER test rows, 48.8%, where the two levels are identical) and checks the row count before writing.

Normalisation for comparison. For leakage checks and scoring alignment, diacritics and punctuation are removed and the alef, ؤ and ي/ي variants are unified. Training text keeps its spelling apart from tashkeel and tatweel.

Leakage. Every training file was checked against both locked test sets and found clean. Matches under five words are reported but not counted, because short formulaic fragments recur in any corpus; BAREC alone repeats 311 sentences between its own train and test splits.

Contamination in a public corpus. 3,190 of Baseet’s 34,294 rows (9.3%) overlapped our locked test sets: 1,829 SAMER test sentences and 1,361 BAREC test sentences. Seen from the other side, 47% of SAMER’s substantive test sentences appear somewhere in Baseet. Training on it unfiltered would have inflated every later score with no visible error. The rows were removed and the remaining 31,104 re-checked: zero matches.

Meaning filter. Applied to the Baseet candidates, the filter keeps 6,888 pairs across the three levels. Its purpose shows at the strongest level: the 5th percentiles of Baseet’s level-1 targets are 0.44 LaBSE similarity and 26% of the source’s length, and 10% of those targets drop a number, against 0.646, 67% and none for DAASI’s human rewrites.

3.2. Exploratory Data Analysis (EDA)

Table 7. Corpus statistics, computed on the files we hold. Identity = target equals source after removing diacritics; length ratio = median of target words over source words.

Corpus	Rows	Median words	≤3 words	Diacritized	Identity	Length ratio
SAMER train	14,343	7	16.4%	0.0%	42.1%	1.00
SAMER test (locked)	3,277	6	23.6%	0.0%	48.8%	1.00
BAREC test (locked)	7,286	10	16.9%	53.8%	—	—
Baseet, strongest level	31,104	14	0.0%	36.5%	0.1%	0.73
DAASI	1,509	12	8.4%	32.3%	0.0%	1.00

Four properties of the corpora (Table 7, Figure 4) explain the behaviour of both models:

- **SAMER carries no diacritics**, while 54% of BAREC test and 36% of Baseet do. A model trained only on SAMER has never seen a diacritized input.
- **SAMER simplifies words, not sentences.** Its median length ratio is exactly 1.00: the editors substituted vocabulary and kept structure. Baseet’s strongest level shortens a 14-word median source to 9 words.

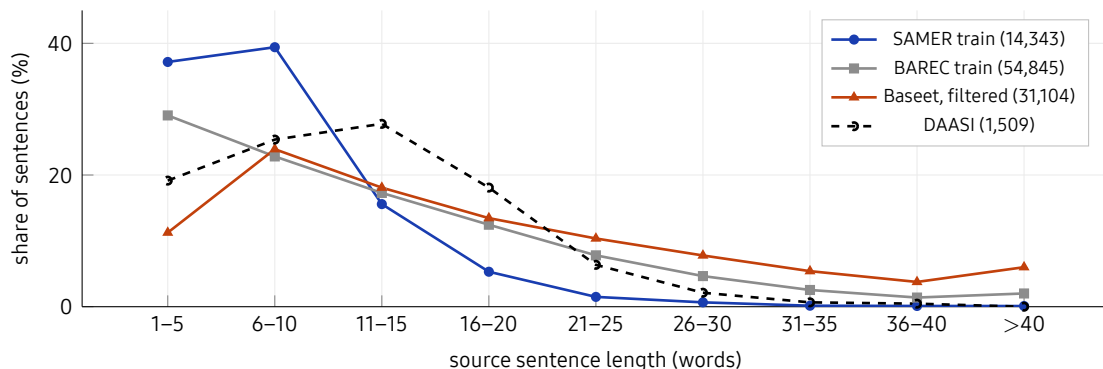


Figure 4. Source sentence length by corpus. SAMER is short literary text; Baseet and BAREC reach the long sentences readers struggle with.

- **No corpus teaches splitting.** A sentence is added in 0% of SAMER pairs, 2–4% of Baseet’s and 8% of DAASI’s, and word length and clitic load barely change in any of them. Existing data can teach shorter, more common words; it cannot teach the restructuring the reading research supports most directly [6].
- **Half of SAMER needs no change.** 42–49% of pairs are identical, and SAMER splits by chapter rather than by novel, so its test set is in-domain. Both make SARI on SAMER read optimistically.

3.3. Modeling

Setup. Reference-based scores use SARI [18] through the `evaluate` implementation, beside BLEU, BERTScore between source and output [19], and the fall in predicted BAREC level. Every score is paired with the copy baseline and split into rows a human changed and rows a human left unchanged. Each final model was run once per test set after training and selection were complete.

RQ1: the reference-based view. Because half of SAMER needs no change, copying scores SARI 77.50 on the full test set but only 56.06 on the 1,678 rows a human changed, and the second number is the bar a model must clear. Table 8 and Figure 5 report model 1, with the harness validated first: it reproduced the training run’s 2,983 dev predictions row for row.

Table 8. Model 1 on test data with human references, scored once with `score.py`. SARI is split by whether the human reference changed the source (DAASI has no unchanged pairs). BERTScore compares output with source; readability drop is the fall in predicted BAREC level (positive = easier). DAASI sources and references have tashkeel stripped.

System	SARI all	SARI changed	SARI unchanged	BLEU	BERT-Score	Read-ability drop	Copy rate
SAMER test (3,277 rows, locked, in domain)							
Copy the input	77.50	56.06	100.00	72.03	1.00	0.00	100.0%
Model 1, greedy	75.88	61.83	90.62	71.68	0.98	0.08	65.3%
Model 1, beam 4	76.29	61.95	91.34	72.22	0.98	0.07	69.4%
DAASI held-out (350 pairs, government and insurance text, out of domain)							
Copy the input	45.05	45.05	—	25.25	1.00	0.00	100.0%
Model 1, greedy	43.87	43.87	—	23.66	0.99	−0.03	77.7%

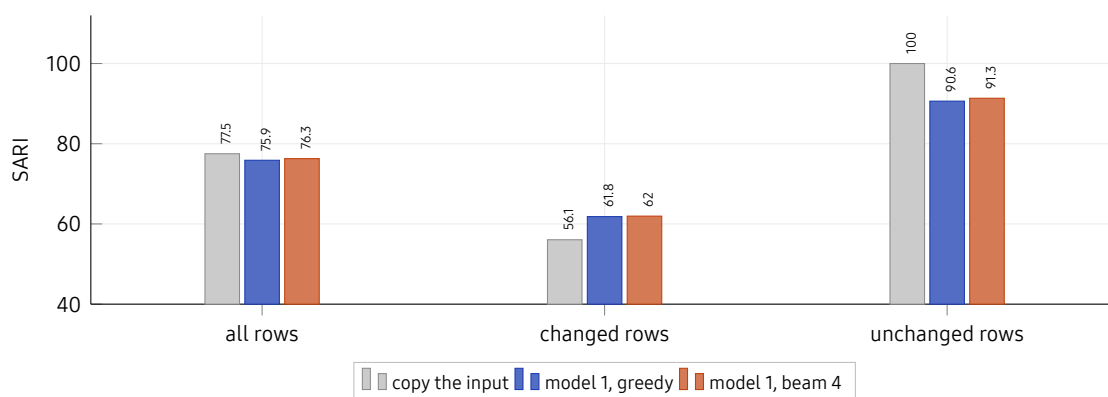


Figure 5. SARI on the locked SAMER test set by reference bucket. Model 1 clears the copy baseline where a human simplified and loses to it where a human left the sentence alone.

Model 1 simplifies where it should and over-edits where it should not. On the 1,678 test rows a human simplified, greedy decoding scores 61.83 against the copy baseline's 56.06 (+5.77 SARI). On the 1,599 rows a human left alone it scores 90.62 against a perfect 100, because it edits 13.6% of sentences that needed no change. The two effects net to -1.62 overall, so the headline number would report a working model as worse than doing nothing. Beam search changes little (61.95 on changed rows).

What model 1 learned. The model returns 65.3% of test rows unchanged where the editors left 48.8%, and leaves 45.3% of the rows an editor rewrote untouched. When it edits, 75.4% of its edits change one word (editors: 49.5%), and its median output-to-input length ratio is 1.00: it substitutes vocabulary and never restructures (Figure 7a). Figure 6 shows the role of checkpoint selection: the SARI that chose the checkpoint had been flat within noise since step 1,800, while the copy rate was still falling.

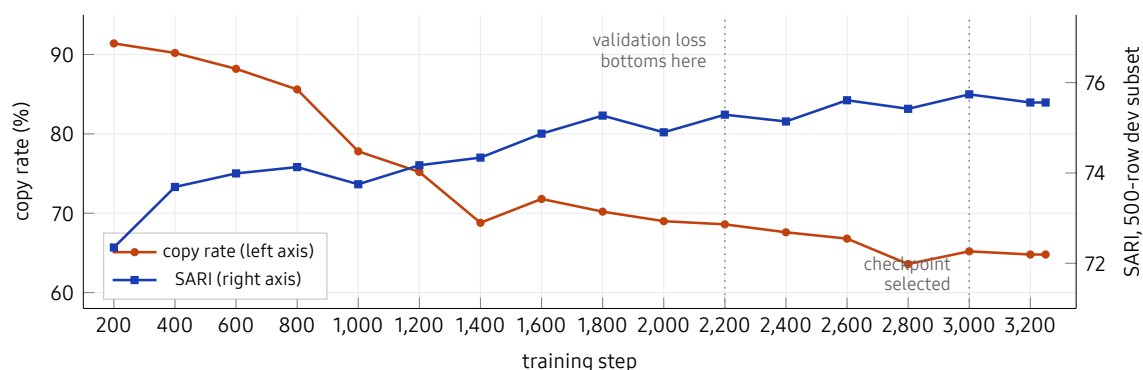


Figure 6. Model 1 training. The checkpoint was selected on a 0.18-point SARI difference while the model's copying behaviour was still changing.

Diacritized and out-of-domain input. On BAREC's 3,918 test sentences that carry tashkeel, raw input keeps 63.7% of the source words and 31.9% of outputs lose more than half the sentence; with tashkeel stripped the same sentences keep 94.6% and 1.8% lose more than half (Table 9, Figure 7b). On DAASI's held-out government and insurance text, model 1 scores 43.87 against a copy baseline of 45.05 (-1.19) and returns 77.7% of sentences unchanged: from SAMER's novels it learned when not to edit, not how to simplify administrative prose.

Meaning errors. On dev, 832 of model 1's substitutions disagree with the editor, 347 of them on words the editor deliberately kept, and some reverse the meaning: عرلة (isolation) became

Table 9. Model 1 on the locked BAREC test set (7,286 sentences, no reference simplifications), greedy decoding. Stripped: tashkeel removed from the source before the model sees it, as the application does.

Input	Copy rate	BERTScore	Readability drop	Source words kept	Rows losing >half
Stripped (as deployed)	68.6%	0.98	-0.04	94.9%	1.4%
Raw, with tashkeel	37.5%	0.91	-0.45	78.2%	17.5%

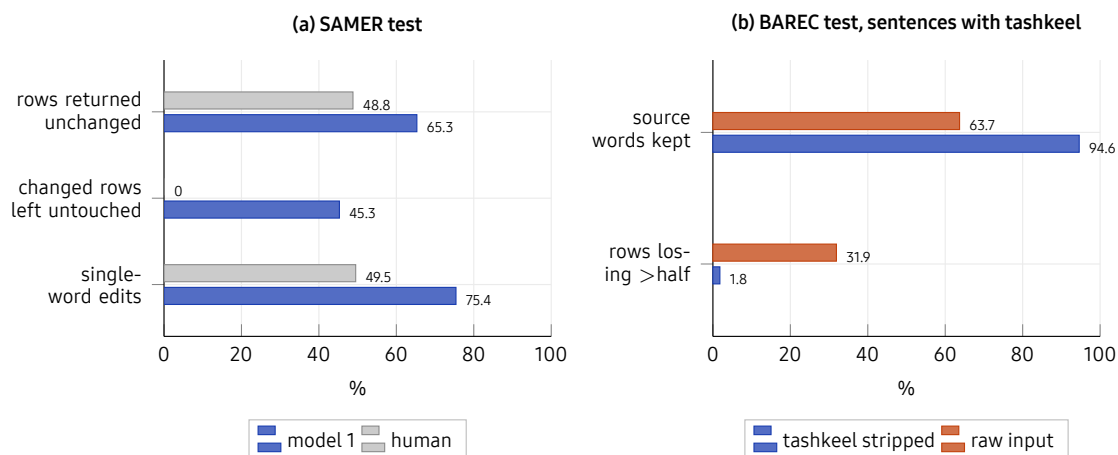


Figure 7. Model 1 behaviour. (a) SAMER test, greedy, against the editors' references. (b) BAREC test, the 3,918 sentences that carry tashkeel, greedy, with and without tashkeel stripped.

استعباد (enslavement), ملجَم (bridled) became مسلح (armed). SARI and BLEU are blind to this class of error, which is why BayanBench scores meaning separately from ease.

Model-2 training. Both runs clear model 1's selection score (Table 10, Figure 8); for reference, model 1 scored 63.66 SARI on SAMER dev's changed rows (all 1,754 of them, a superset of the 600 used here). AraT5v2 was still improving at its last evaluation; AraBART peaked at epoch 9 and had been flat since epoch 6. Copying falls from 80–91% of rows to about 51% for both, against 33.3% for the editors on the same rows: the recipe reduces over-conservatism but does not remove it, and AraT5v2 still returns 186 of the 600 changed rows unchanged.

Table 10. Model-2 runs, one T4 GPU each. SARI and copy rate on the 900-row selection sample (600 changed, 300 unchanged SAMER dev rows, tag [s0]). Speed: sentences per second on SAMER test sources, fp16, batch 64, greedy.

	AraT5v2	AraBART
Parameters	368M	139M
Training time	6,783 s	2,600 s
Best checkpoint (step of 14,210)	14,200	12,780
SARI, 600 changed rows	66.58	64.71
SARI, all 900 rows	75.64	73.62
Copy rate, 900 rows (editors: 33.3%)	51.0%	51.4%
Generation speed (sentences/s)	80.8 [†]	292.8
[†] measured on model 1, the same architecture and size, in the same run.		

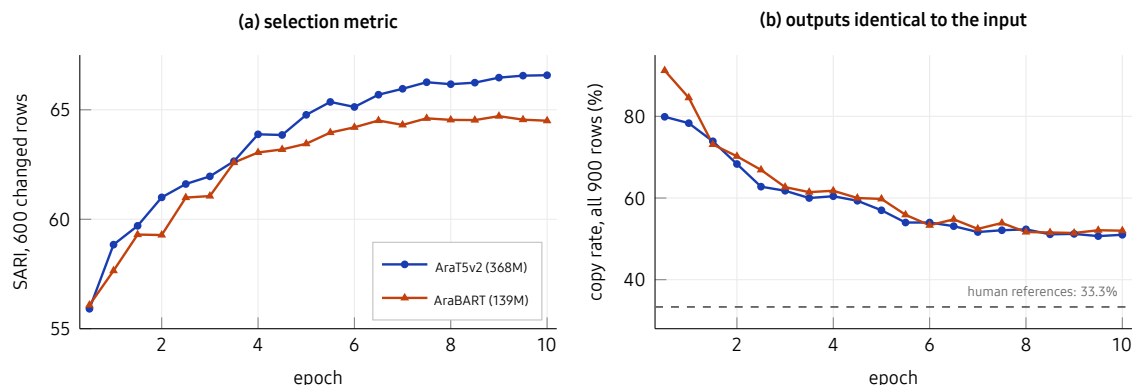


Figure 8. Model-2 training, from the committed notebooks: (a) the selection metric at each of the 20 evaluations; (b) the share of outputs identical to the input, with the editors’ rate as a dashed line.

3.4. Human Evaluation

Agreement between raters. On the 80 outputs given to every rater, the raters agreed moderately on meaning (Krippendorff’s $\alpha = 0.51$, interval metric; 74% of rater pairs gave the same answer) and weakly on Arabic ($\alpha = 0.17$) and ease ($\alpha = 0.19$). The Arabic figure is low partly because almost every output was rated correct, so the raters agreed on 87% of pairs yet the few errors they flagged rarely coincided. Meaning is therefore the one human judgement firm enough to check an automatic scorer against; ease is reported per system but not used as a target.

The meaning scorer against people. Table 11 and Figure 9 compare the scorer’s P(same meaning) with the raters’ verdict on all 245 rated outputs. It separates the outputs the raters kept from the rest with an AUC of 0.85 [0.78–0.91], rising to 0.92 on the 80 outputs with three or more raters, where the human verdict is itself more reliable; it is 0.91 when a minor change counts as kept. Its scores are bimodal, so the cut-off matters less than the ranking. As a pass/fail rule at 0.5 it catches 42 of the 51 outputs the raters judged changed, including all 11 with a major change, and fails 53 of the 194 that the raters kept: the scorer is stricter than people, so “meaning kept” is a conservative rate. A cut-off of 0.3 would have failed seven fewer kept outputs and caught the same losses; we kept the 0.5 set before the ratings rather than tune the threshold on the sample used to check it.

Table 11. The open meaning scorer (Gemma 4 31B, P(same meaning)) against the human verdicts on the 245 rated single outputs. AUC: the chance that an output the raters kept scores higher than one they did not (0.5 = chance, 1 = perfect); 95% intervals from a bootstrap over documents.

Human verdict counted as “meaning kept”	Kept	AUC [95% CI]
Majority “same meaning”	194 of 245	0.85 [0.78–0.91]
Majority not “major change” (a minor change counts as kept)	234 of 245	0.91 [0.86–0.96]
Every rater “same meaning”	175 of 245	0.83 [0.76–0.89]
Majority “same meaning”, tasks with three or more raters	64 of 80	0.92 [0.86–0.97]
The scorer as a pass/fail rule (kept = majority “same meaning”)		
Threshold on P(same)	Losses caught	Kept outputs failed
0.3	42 of 51	46 of 194
0.5 (frozen)	42 of 51	53 of 194
0.7	45 of 51	61 of 194

Each system, as people rated it. Table 12 and Figure 10 show each system on the 35 changed

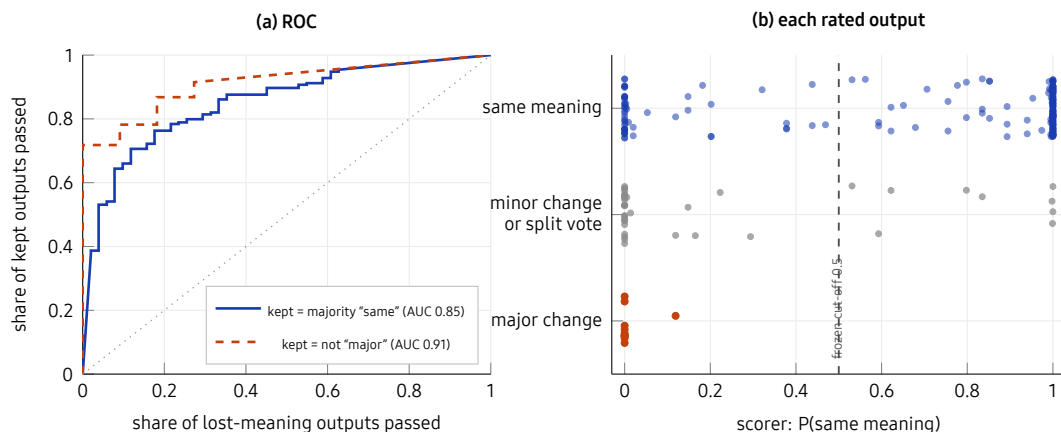


Figure 9. The meaning scorer against the raters, on 245 rated dev outputs. (a) ROC curves for two definitions of a kept meaning; the dotted diagonal is chance. (b) Every rated output by the raters' majority answer (194 same, 40 minor or split, 11 major) against the scorer's P(same); rows are jittered vertically for legibility.

outputs drawn evenly from it. The raters kept the meaning in 94% of the int8 AraT5v2 bundle's rewrites and 89% of the untagged AraT5v2's, against 71% for each of the three AraBART systems. The scorer, applied to the same outputs, also places every AraT5v2 system above every AraBART one, and is 6 to 22 points stricter than the raters on each system. The raters judged about half of every system's rewrites easier (49–57%) and 89–97% of them correct Arabic. Because copies were not rated, these are rates among outputs that changed, not over all inputs as in subsection 3.5.

Table 12. Each system as the raters and the scorer saw the same outputs: the 35 changed outputs per system drawn evenly (copies of the input were not rated). Rates in %, a strict majority of the task's raters; 95% intervals from a bootstrap over documents. *Scorer*: $P(\text{same}) \geq 0.5$ on the same outputs.

System	Same meaning: raters	Same meaning: scorer	Easier: raters	Correct Arabic: raters
AraT5v2, int8 app	94 [85–100]	80 [67–92]	49 [30–67]	91 [82–100]
AraT5v2, no tag	89 [78–97]	83 [71–94]	54 [36–72]	89 [77–97]
AraT5v2, [s2]	77 [62–91]	63 [46–78]	49 [34–65]	97 [90–100]
AraBART, int8 app	71 [56–86]	51 [31–72]	54 [38–70]	94 [86–100]
AraBART, no tag	71 [52–88]	54 [38–71]	51 [33–68]	91 [82–100]
AraBART, [s2]	71 [57–86]	49 [34–63]	57 [43–72]	97 [92–100]

The reader with dyslexia. Table 13 gives the sixth rater's answers. Of 30 rewrites, 17 were easier to read for them, 9 no different and 4 harder; their answer matched the team's majority on 67% of the shared outputs, about as often as two team raters agreed with each other on ease (61%). The int8 AraT5v2 bundle, the most cautious system, was the only one never found easier. One reader is not a study, but the direction matches the benchmark: the system that keeps meaning best is also the one that simplifies least.

Side-by-side comparisons. Table 14 counts the team's votes. The clearest contrast is architecture without a tag, where AraT5v2 was preferred in most votes; the other contrasts hold too few votes to separate the systems.

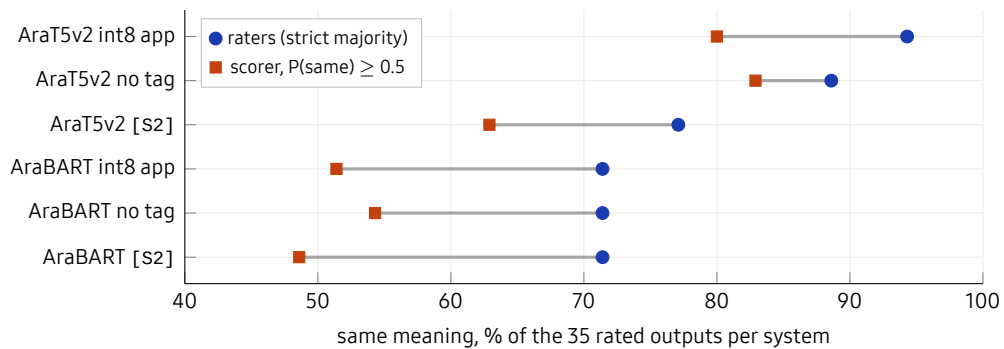


Figure 10. Same meaning, per system, on the same 35 outputs each: the raters’ verdict (circle) and the scorer at $P(\text{same}) \geq 0.5$ (square).

Table 13. The reader with dyslexia: “was it easier for you to read?” on 30 outputs (five per system), against the original.

System	Easier	Same	Harder
AraT5v2, int8 app	0	3	2
AraT5v2, no tag	4	0	1
AraT5v2, [S2]	3	2	0
AraBART, int8 app	3	2	0
AraBART, no tag	5	0	0
AraBART, [S2]	2	2	1
All	17	9	4

Table 14. Side-by-side comparisons by the team raters: which version is better for a reader with dyslexia (easier, with the same meaning). Counts of rater votes.

Contrast	Systems (first vs second)	First	Tie	Second
app	AraBART, int8 app vs AraBART, [S2]	4	4	3
app	AraT5v2, int8 app vs AraT5v2, [S2]	6	2	8
architecture	AraBART, [S2] vs AraT5v2, [S2]	5	5	5
architecture	AraBART, no tag vs AraT5v2, no tag	2	1	10
tag	AraBART, [S2] vs AraBART, no tag	10	0	6
tag	AraT5v2, [S2] vs AraT5v2, no tag	3	7	6

3.5. BayanBench Results

RQ2: model 2 on BayanBench. Each of the seven systems was run once on the BayanBench splits: model 2 on both architectures in the medium setting ([S2]) and without a tag, the two int8 bundles exactly as the app calls them, and copying the input. Table 15 reports the dev split on core items; Table 16 isolates one change at a time, and Table 19 gives the other split. Five findings follow.

Table 15. Model 2 on BayanBench v2, **dev** split, *core* items (380 of 731 items; 129 documents in the split), each system run once. Meaning: the open scorer, Gemma 4 31B; a copy of the input keeps its meaning by definition. Simpler: mean change from selection to output, over the items where there is something to simplify. Rates in %, means with their sign; 95% intervals from a bootstrap over whole documents in brackets. The two [S2] columns send the medium strength tag the model was trained with; the two *int8 app* columns are the quantised bundles the app downloads, which send no tag. All outputs pass through the app's guards. Bold: the best model on each row. *n* = items the measure applies to.

Measure	Copy the input	AraT5v2, [S2]	AraBART, [S2]	AraT5v2, int8 app	AraBART, int8 app	<i>n</i>
Meaning						
Meaning kept (P(same) $\geq$ 0.5, numbers kept), %	100.0	62.5 [57–68]	41.9 [36–49]	78.4 [73–83]	54.7 [49–61]	320
Mean P(same meaning)	1.00	0.63 [0.58, 0.68]	0.42 [0.36, 0.49]	0.79 [0.74, 0.84]	0.55 [0.49, 0.61]	320
Simpler (mean change)						
Longest clause: words cut (clause over 15)	+0.0	+6.4 [+5.2, +7.6]	+6.0 [+5.0, +6.9]	+1.4 [+0.6, +2.3]	+3.8 [+2.8, +4.7]	106
Reading level: levels lower (level 10+)	+0.00	+0.20 [+0.07, +0.35]	+0.25 [+0.11, +0.38]	+0.01 [-0.04, +0.07]	+0.21 [+0.10, +0.31]	255
Hard words removed (items with any)	+0.00	+0.58 [+0.33, +0.83]	+0.46 [+0.28, +0.66]	+0.34 [+0.19, +0.49]	+0.47 [+0.29, +0.65]	148
Checks, %						
Every number kept	100.0	76.0 [59–91]	72.0 [56–86]	72.0 [50–88]	72.0 [52–88]	25
No house-rule break	100.0	99.7 [99–100]	99.7 [99–100]	100.0	99.7 [99–100]	320
Behaviour, %						
Easy text left unchanged (C)	100.0	19.1 [8–29]	12.8 [5–21]	40.4 [26–57]	21.3 [11–31]	47
Cut-off selection keeps its ending (E1)	100.0	36.7 [24–49]	44.9 [32–57]	49.0 [38–59]	55.1 [42–67]	49
Protected text: letters unchanged (F)	100.0	11.7 [4–26]	6.7 [0–16]	48.3 [21–91]	26.7 [11–55]	60
Flags (not failures), %						
Negation count changed	0.0	20.4 [9–31]	18.0 [8–28]	2.1 [0–6]	2.1 [0–6]	47

F1 Meaning and simplification trade off. The medium setting cuts 6.4 (AraT5v2) and 6.0 (AraBART) words from the longest clause and keeps the meaning in 62.5% and 41.9% of outputs. Without the tag the same models cut 3.7 and 3.3 words and keep 77.8% and 55.6%; the int8 AraT5v2 bundle cuts 1.4 and keeps 78.4% (Figure 11). Every system that simplifies more keeps meaning less often, and none reaches both.

F2 AraT5v2 keeps meaning better at every setting. At the same setting AraT5v2 keeps the meaning 22.2 points more often than AraBART without a tag [+16.6, +27.5], 20.6 with the

[s2] tag and 23.8 as an int8 bundle, every interval excluding zero, while the two cut long clauses by similar amounts without a tag. The human ratings agree (Figure 10).

F3 Simplification is mostly structural. The predicted reading level falls by at most 0.25 levels on average, and each output removes 0.34–0.58 hard words on average, where a clause loses up to 6.4 words: the models shorten clauses far more than they change vocabulary.

F4 Restraint is weak. Easy text comes back unchanged in only 12.8–40.4% of C items, protected text keeps its letters in 6.7–48.3%, and a selection cut mid-sentence keeps its ending in 36.7–59.2% of E1 items; the rest are completed, sometimes with invented words (Table 18). No training corpus shows these behaviours: SAMER’s unchanged pairs are literary sentences, and no corpus contains protected or cut-off text. Numbers survive in 72.0–76.0% of the core items that contain one.

F5 The deployed format buys meaning with ease. The bundles the app ships send no strength tag (Table 16). Dropping the tag raises meaning kept by 15.3 (AraT5v2) and 13.8 (AraBART) points and cuts 2.7 and 2.6 fewer words from the longest clause. Quantising to int8 leaves meaning unchanged for both models; for AraT5v2 it also moves the model toward copying (2.3 fewer words cut, 14.9 points more easy text left alone), which makes the shipped AraT5v2 bundle the safest system and the least simplifying one.

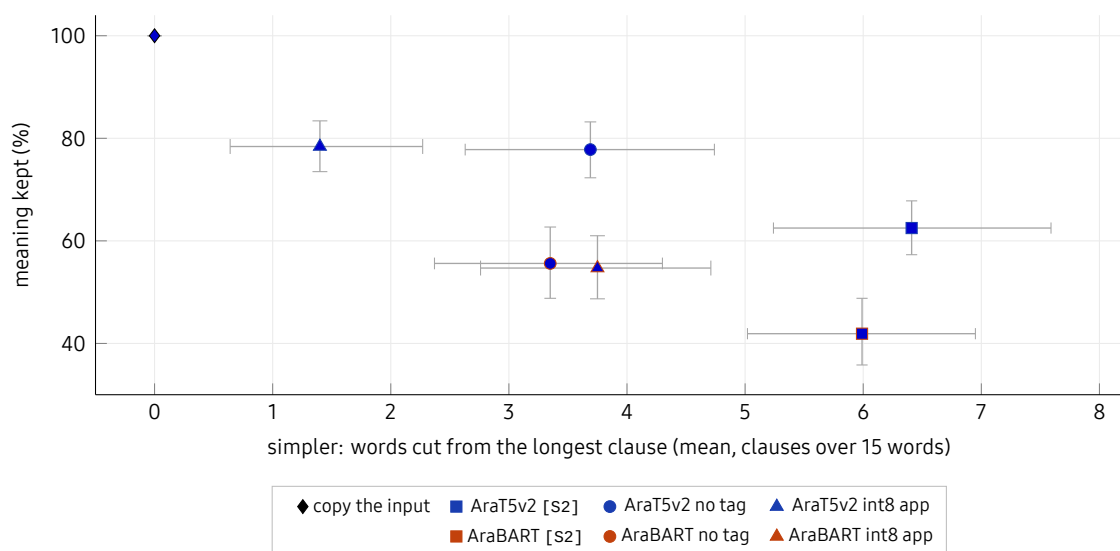


Figure 11. Meaning kept against simplification, dev split, core items, with 95% document-bootstrap intervals. Copying the input sits at the top left by definition. Colour: the model; shape: the setting.

Across item sets. Table 17 reports the four item sets side by side. Meaning is kept at similar rates on text published after every corpus we use (outside) as on core, so the scorer’s view does not depend on BAREC’s genres; the gains in clause length are smaller away from core, and restraint is weak everywhere.

Error analysis. Table 18 shows test outputs of AraT5v2 in the medium setting. The first is the intended behaviour: a 17-word sentence split into two cause-linked clauses. The second passes every code check and reverses a quantity, which is why meaning needs a scorer and not code alone. The others are the failures of F4. One example is not printed: in a Qur’anic verse (*al-Kāfirūn* 109:4) the model dropped the negation **لَا**, inverting the verse’s meaning, the most serious error we observed and the reason protected text must never reach the model.

Table 16. Paired effects on the dev split (core items): one change at a time, same items. Differences in points (rates) or words (clause), with 95% document-bootstrap intervals; * = the interval excludes zero.

One change	Meaning kept (points)	Words cut from the longest clause	Easy text unchanged (points)
AraT5v2: drop the [s2] tag	+15.3* [+10.7, +20.7]	-2.7* [-3.6, -1.9]	+6.4 [+0.0, +14.3]
AraBART: drop the [s2] tag	+13.8* [+6.2, +21.2]	-2.6* [-3.4, -1.8]	+8.5 [+0.0, +18.0]
AraT5v2: PyTorch → int8 app	+0.6 [-3.1, +4.5]	-2.3* [-3.1, -1.4]	+14.9* [+4.1, +29.4]
AraBART: PyTorch → int8 app	-0.9 [-3.7, +2.0]	+0.4* [+0.0, +0.9]	+0.0 [-6.8, +6.0]
[s2]: AraBART → AraT5v2	+20.6* [+13.4, +27.4]	+0.4 [-0.6, +1.4]	+6.4 [+0.0, +12.7]
No tag: AraBART → AraT5v2	+22.2* [+16.6, +27.5]	+0.3 [-0.9, +1.4]	+4.3 [-9.3, +17.1]
int8 app: AraBART → AraT5v2	+23.8* [+18.1, +29.7]	-2.4* [-3.6, -1.1]	+19.1* [+2.1, +38.2]

Table 17. The four item sets on the dev split, reported side by side and never averaged; number of items in brackets. *Core*: BAREC test documents; *held-out*: news and legal text; *outside*: web text published after every corpus we use; *written*: medicine-leaflet sentences written for the benchmark.

System and measure	Core	Held-out	Outside	Written
AraT5v2, [s2]				
Meaning kept, %	62.5 (320)	69.0 (184)	57.9 (140)	63.0 (27)
Longest clause: words cut	+6.4 (106)	+6.2 (37)	+3.6 (36)	+4.0 (3)
Easy text left unchanged, %	19.1 (47)	8.6 (58)	16.7 (24)	—
AraBART, [s2]				
Meaning kept, %	41.9 (320)	53.8 (184)	47.1 (140)	37.0 (27)
Longest clause: words cut	+6.0 (106)	+6.0 (37)	+6.0 (36)	+2.7 (3)
Easy text left unchanged, %	12.8 (47)	10.3 (58)	25.0 (24)	—
AraT5v2, int8 app				
Meaning kept, %	78.4 (320)	86.4 (184)	74.3 (140)	85.2 (27)
Longest clause: words cut	+1.4 (106)	+2.4 (37)	+2.1 (36)	+1.7 (3)
Easy text left unchanged, %	40.4 (47)	31.0 (58)	45.8 (24)	—
AraBART, int8 app				
Meaning kept, %	54.7 (320)	66.3 (184)	52.1 (140)	51.9 (27)
Longest clause: words cut	+3.8 (106)	+3.1 (37)	+4.8 (36)	+3.3 (3)
Easy text left unchanged, %	21.3 (47)	24.1 (58)	25.0 (24)	—

Table 18. Test outputs of AraT5v2 ([s2]), from the BayanBench test split. Glosses are ours.

Track	Source	Output	What happened
B	فالمهم الصفح والتسامح وذلك للتخفيف على الصحة النفسية للفرد ولا انتشار المحبة والتعاون بين أفراد المجتمع بشكل عام.	فالمهم الصفح والتسامح؛ لأن ذلك يساعد على تحسين الصحة النفسية للفرد، ويساعد على انتشار المحبة والتعاون بين أفراد المجتمع بشكل عام.	Split into two clauses joined by a stated cause; meaning kept.
B	منذ قرنين على الأكثر بدأ الإنسان يدخل دائرة حلزونية متصاعدة متفاقمة بين الطلب والاستجابة له ...	منذ قرنين على الأقل، بدأ الإنسان يدخل دائرة حلزونية متصاعدة بين الطلب والاستجابة له ...	“At most two centuries” became “at least”: a reversal every code check passes.
C	يعتبر منصب الرئيس شاغرا في أي من الحالات التالية:	يعتبر منصب الرئيس شاغرا في أي حالة:	Easy legal text edited; “in any of the following cases” lost its list.
E1	الشكل المنطقي الأساسي في علم المنطق، ويتم تحديد صحة المفاهيم من خلال صحة المفاهيم.	الشكل المنطقي الأساسي في علم المنطق، ويتم تحديد صحة المفاهيم من خلال اختبار صحة المفاهيم.	A cut-off selection completed with invented words and a full stop.
A2	... بنتيجة (١/٣) في المباراة ...	... بنتيجة (١) في المباراة ...	Persian-range digits dropped.

Table 19. Model 2 on BayanBench v2, **test** split, **core** items (749 of 1250 items; 197 documents in the split), each system run once. Meaning: the open scorer, Gemma 4 31B; a copy of the input keeps its meaning by definition. Simpler: mean change from selection to output, over the items where there is something to simplify. Rates in %, means with their sign; 95% intervals from a bootstrap over whole documents in brackets. The two [s2] columns send the medium strength tag the model was trained with; the two *int8 app* columns are the quantised bundles the app downloads, which send no tag. All outputs pass through the app’s guards. Bold: the best model on each row. *n* = items the measure applies to. The meaning rows are left out: this split’s outputs are not yet scored for meaning.

Measure	Copy the input	AraT5v2, [s2]	AraBART, [s2]	AraT5v2, int8 app	AraBART, int8 app	<i>n</i>
Simpler (mean change)						
Longest clause: words cut (clause over 15)	+0.0	+6.3 [+5.0, +7.9]	+6.7 [+5.4, +8.3]	+1.9 [+1.0, +2.9]	+4.8 [+3.5, +6.1]	212
Reading level: levels lower (level 10+)	+0.00	+0.40 [+0.31, +0.49]	+0.38 [+0.30, +0.47]	+0.14 [+0.08, +0.20]	+0.28 [+0.19, +0.38]	543
Hard words removed (items with any)	+0.00	+0.61 [+0.46, +0.81]	+0.68 [+0.54, +0.85]	+0.31 [+0.22, +0.44]	+0.59 [+0.46, +0.75]	328
Checks, %						
Every number kept	100.0	82.7 [65–95]	78.7 [63–90]	84.0 [70–94]	76.0 [59–88]	75
No house-rule break	100.0	99.8 [99–100]	99.7 [99–100]	100.0	100.0	659
Behaviour, %						
Easy text left unchanged (C)	100.0	14.3 [7–22]	11.7 [5–19]	48.1 [36–61]	24.7 [15–35]	77
Cut-off selection keeps its ending (E1)	100.0	36.5 [25–47]	38.8 [27–50]	38.8 [28–49]	45.9 [34–58]	85
Protected text: letters unchanged (F)	100.0	4.4 [0–10]	4.4 [0–11]	47.8 [30–70]	24.4 [16–36]	90
Flags (not failures), %						
Negation count changed	0.0	23.1 [17–30]	24.1 [17–32]	7.1 [4–11]	18.4 [13–25]	139

3.6. On the Device

RQ3: cost on the phone. Both simplifiers were exported to ONNX with the tokenizer inside the bundle and quantised to int8 (Table 20). The AraBART bundle fits the 250 MB budget; the AraT5v2 bundle does not, because 46% of its weights are two untied 110k-row embedding matrices. We tested both bundles in the app on a Xiaomi Mi 11X (Snapdragon 870, Android 13, 6 GB RAM), a phone that now sits in the mid-range, with the same two-sentence paragraph and six cold starts each. Because output streams, the reader waits for the first token, not the whole sentence: with AraBART 82–172 ms for a paragraph’s first sentence and 28–30 ms for each later one, with AraT5v2 177–318 ms and 55–85 ms. AraBART loads in 1.0–1.7 s against 2.6–7.3 s, and finishes a sentence about twice as fast, the same ratio as on the laptop. Quantisation leaves meaning unchanged for both models and shifts AraT5v2 toward copying (F5).

Table 20. On-device cost. Laptop: 4 CPU threads, median and 90th percentile per BayanBench test sentence, the app’s own decoding loop. Phone: Xiaomi Mi 11X (Snapdragon 870, Android 13, 6 GB RAM), from the app’s logs over six cold starts each on the same two-sentence paragraph (AraT5v2 on 26 September, AraBART on 3 October).

	AraBART, int8	AraT5v2, int8
Download (encoder + decoder + tokenizer)	222 MB	471 MB
Laptop CPU, per sentence (median / p90)	0.10 / 0.24 s	0.20 / 0.43 s
Phone: model load, cold	1.0–1.7 s	2.6–7.3 s
Phone: first token, first sentence	82–172 ms	177–318 ms
Phone: first token, later sentences	28–30 ms	55–85 ms
Phone: whole sentence	0.23–1.10 s	0.48–2.24 s
Read-aloud voice (Nabra-7M-Distill, int8)	8.7 MB	

Diacritization. Four systems were benchmarked on the 743-sentence Tashkeela test fixture (Table 21). Libtashkeel was selected: CATT ties it on accuracy (0.01 points, about six characters in 67,576) but is 16 times larger and carries a non-commercial licence, and CAMEL Tools changed the base letters of 32 of 100 sentences, corrupting the text it was asked to vocalise.

Table 21. Diacritization benchmark. DER = diacritic error rate; base-letter corruption measured on 100 sentences.

System	DER	DER, no case	WER	Base letters changed	Size	Licence
CAMEL Tools MLE [39]	16.57%	13.97%	56.73%	32%	129 MB	MIT/GPL
CATT [40]	8.44%	6.92%	29.40%	0%	77.9 MB	CC BY-NC
Mishkal [41]	12.74%	10.12%	39.33%	0%	4.3 MB	GPLv3
Libtashkeel [42]	8.43%	6.91%	29.57%	0%	4.8 MB	MIT

3.7. User Interface

Figure 12 shows the interface on the phone. Selecting Arabic text in any app offers تبسيط; choosing it opens the result over the app, the rewrite appearing sentence by sentence as it is decoded. One toggle switches between *Simplified* and *Original*; below it are read-aloud, copy and share, and the current word is highlighted while Bayan reads. The panel can float over the page or shrink to a bar, so the reader keeps the host app in view while listening. The app follows Android’s Material 3 design rather than a design system of its own, and sets Arabic right to left without justification, because justification in Arabic stretches words and opens uneven gaps. Settings hold the reader’s typeface, size (22 sp by default), line height (1.8) and word spacing,

the one spacing adjustment that carries over to cursive Arabic, together with the background, the model, the voice and its speed, and word highlighting.

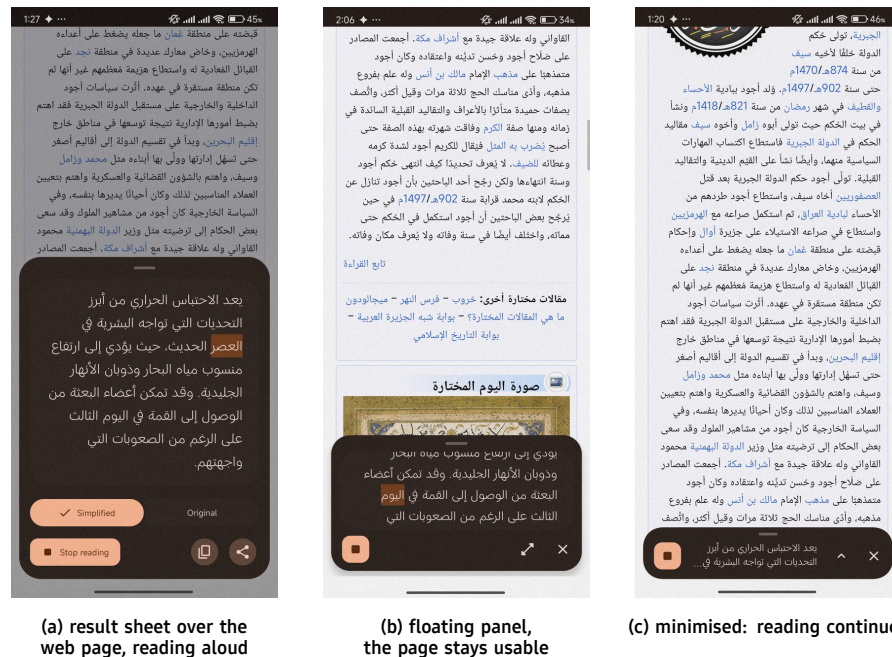


Figure 12. The app on the phone, over an Arabic Wikipedia page. (a) The result sheet, reading aloud with the current word highlighted. (b) The floating panel. (c) Minimised to a bar while reading continues.

3.8. Testing and Improvements

Testing. The first plugin shell shipped 41 unit and Robolectric tests. The app tested on the phone unit-tests its text pipeline (sentence splitting, punctuation repair, clitic normalisation and the retention fallback), and its download pipeline checks every file by SHA-256. The leakage checker is tested against a planted duplicate and must exit non-zero. The benchmark package has 12 tests on a built-in toy dataset, and its prediction command reproduces our model-2 outputs on all 731 dev items, as an independent rerun by a teammate confirmed. Running the pipeline end to end on real data, not fixtures, exposed four harness faults: the scoring script put every BAREC row in the “unchanged” bucket; the shared prediction script omitted model 1’s training prefix, which alone cost twelve SARI points; length-grouped batching shuffled the evaluation set; and under transformers 5 the same models produced garbled outputs, because AraT5v2’s tied embeddings and AraBART’s sentence markers are handled differently. Each now has a guard; the last makes the prediction command refuse to run under version 5.

Improvements. Each measured problem produced a change (Table 22); the findings of the final evaluation are discussed in subsection 4.2.

Table 22. What the measurements changed.

Finding	Change
Copying scores SARI 77.5 on SAMER test	Copy baseline and changed/unchanged split reported with every score
9.3% of Baseet overlaps our test sets	Rows removed; leakage check required for every training file
Model 1 collapses on diacritized input	Tashkeel stripped before the simplifier, in training and in the app
Model 1 chosen on SARI noise over a copy-heavy dev set	Model 2 selects on changed-row SARI only
Model 1 over-edits unchanged rows yet copies 65% of all rows	All unchanged pairs kept in model-2 training
SAMER teaches word substitution; model 1 below copying on DAASI	DAASI and meaning-filtered Baseet added, each under its own tag
Baseet's LLM targets often summarise	Meaning filter calibrated on human rewrites
Prediction script omitted the training prefix	Prefix a required setting of every model and bundle
Length grouping shuffled the evaluation set	Grouping off; alignment check stops the run
SARI is blind to meaning errors	BayanBench: meaning scored separately from ease, with code checks
The closed judge was quota-bound, pass/fail and not rerunnable	Open meaning scorer, continuous, rerun by anyone, checked against six raters
Under transformers 5 the models' outputs changed silently	Prediction refuses to run under version 5
The model invents text around headings and fragments	Sentences under six words pass through unchanged in the app

4. Projected Impact

4.1. Accomplishments and Benefits

The answers to the research questions are these. **RQ1:** no. On copy-heavy test sets, SARI rewards inaction and is blind to reversals of meaning; a model that beats copying on the sentences editors changed still scores below copying overall, and what it learned from SAMER is vocabulary substitution without restructuring. An open meaning scorer, by contrast, agrees with human raters (AUC 0.85) and can be rerun by anyone. **RQ2:** partly. Model 2 shortens long clauses in every setting, and AraT5v2 keeps the meaning of 62.5–78.4% of outputs, clearly more than AraBART; but every setting trades meaning for simplification, the safest one simplifies least, and no setting leaves easy text alone. **RQ3:** yes, with one caveat. Streamed output reaches the reader in under a third of a second on a mid-range phone, and quantisation costs neither model measurable meaning; the int8 AraBART bundle fits the 250 MB budget, while the AraT5v2 bundle, the more faithful of the two, needs 471 MB.

Benefits. For readers: help where and when the difficulty occurs, offline and private, with the original always one tap away. For schools: a tool that runs on the phones students already own. For the field: a documented case of test-set contamination in a public Arabic simplification corpus, an evaluation protocol that exposes copy-heavy scores, and a benchmark whose meaning scorer has been checked against people, measuring what an Arabic reader with dyslexia needs rather than agreement with one editor.

Table 23. Objectives from subsection 1.2, against the evidence.

Objective	Evidence	Outcome
O1 Beat copying on changed rows; easier text without losing meaning	Model 1: 61.83 vs 56.06 on SAMER test (+5.77). Model 2: 66.58 on changed dev rows; on BayanBench, meaning kept in 41.9–78.4% of outputs while cutting 1.4–6.4 words from the longest clause, and raters kept the meaning of 71–94% of changed outputs; restraint not met	partly met
O2 Diacritize without changing letters	Libtashkeel: DER 6.91% without case endings, 0 of 100 sentences altered, 4.8 MB; not in the tested build	partly met
O3 Run on a phone within 250 MB	Both bundles run on a Snapdragon 870 phone, first token in 28–318 ms; AraBART int8 is 222 MB, AraT5v2 int8 471 MB	partly met
O4 Evidence from readers	A reader with dyslexia rated 30 rewrites and 10 comparisons with informed consent: 17 easier, 4 harder; no reading-time or comprehension study	partly met

4.2. Future Improvements

The evaluation points to specific next steps. The central one is to move the trade-off of F1: a model that simplifies as much as the tagged setting while keeping meaning as often as the shipped bundle. The meaning scorer, now checked against people, can rank candidate rewrites and select checkpoints, so that training is steered by the behaviours readers need rather than by SARI. Restraint and protection belong in the app as much as in the model: scripture and poetry should be detected and never sent to it, a cut-off selection should keep its ending, and Persian-range digits should be mapped before decoding and restored after. The AraT5v2 bundle can meet the size budget by pruning its vocabulary to the pieces Arabic text uses: 46% of its parameters sit in two embedding matrices, and pruned int8 weights would take about 223 MB. Sentence splitting needs training data that teaches it, which no human-written corpus provides. Last, a reading study with more readers with dyslexia, measuring reading time and comprehension, would turn ratings into evidence about reading.

4.3. Limitations

- **One reader with dyslexia.** The reader rated rewrites on a web page; nobody with dyslexia has yet used the app or been measured reading with it, and the five team raters do not have dyslexia.
- **The meaning scorer is a model.** It agrees with people at AUC 0.85 but is stricter than them, failing 53 of the 194 outputs they kept, so “meaning kept” is a conservative rate. It was checked on dev outputs of the six model-2 systems, from 64 documents; a very different system could be judged differently.
- **Ease and Arabic are weakly agreed.** Raters agreed on ease and on correct Arabic with α of only 0.19 and 0.17, so the benchmark measures ease through clause length, reading level and hard words, and checks Arabic only through house rules.
- **The clause measure credits commas.** A rewrite can shorten its longest clause by adding commas as well as by splitting sentences, which is why it is always read next to meaning.
- **No reference-based test score for model 2.** Model 2 is compared with published work only through its dev selection score; its test evaluation is BayanBench, which has no human references.

- **Some tracks are small.** E3 and E4 hold 64 and 40 test items, and few core items contain a number (Table 15), so these rates separate only large differences.
- **One phone.** On-device figures come from one mid-range phone, and the more faithful AraT5v2 bundle, at 471 MB, exceeds the 250 MB objective.
- **One run per configuration.** Each model was trained once, with one seed, so the comparison between architectures carries no estimate of training variance.

4.4. Conclusion

Bayan shows that an Arabic reading aid for readers with dyslexia can run entirely on a mid-range phone, from inside any app, at interactive latency. Getting there depended less on model size than on evaluation. The field's standard score hid both what a model had learned and what it broke. A public corpus overlapped the test sets. And the behaviours a reader needs — shorter clauses, familiar words, the same meaning, and restraint where the text is already easy — had to be measured one by one before they could be improved. BayanBench makes those measurements reproducible by anyone on the team, its meaning scorer agrees with human raters, and its results set the next goal plainly: simplify as much as the boldest setting while keeping meaning as well as the safest.

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5. Team Member Review and Comment

<ATTACH A TEAM PICTURE HERE>

NAME	REVIEW and COMMENT
Marwan Elamami	
Abdulrahman Khengari	
Ahmed Alaeb	
Sanad Ali	
Mohammed Thabet	
Abdul Majid Mraied	

6. Instructor Review and Comment

CATEGORY	SCORE	REVIEW and COMMENT
IDEA	__/10	
APPLICATION	__/30	
RESULT	__/30	
PROJECT MANAGEMENT	__/10	
PRESENTATION & REPORT	__/20	
TOTAL	__/100	